

Matthew Champagne

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EDUCATION

Stony Brook University

B.E Computer Engineering;

Stony Brook, NY

Aug 2019 – May 2023

Stony Brook University

M.S Computer Engineering;

Stony Brook, NY

Aug 2023 – May 2025

WORK EXPERIENCE

General Dynamics Mission Systems

Senior Software Engineer

Orlando, FL, USA

September 2025 – Present

- Develop embedded C software on Zephyr RTOS for Army training systems.
- Support player units and MILES detectors for tactical engagement tracking.
- Work across GPS, NFC, LTE, and BLE subsystems.

Orbit International Corporation

Junior Embedded Systems Design Engineer

Hauppauge, NY, USA

Jan 2023 – September 2025

- Lead the software development effort for multiple projects including new and legacy products.
- Setup, documented, and maintained a self-hosted GitLab server for Orbit Instrument's Engineering department.
- Helped migrate our sister company from SVN to GitLab.

WORK PROJECTS

General Dynamics WTA/UAS Training System (C on Zephyr RTOS, Nordic)

- Develop embedded software in C using Zephyr RTOS for WTA/UAS devices used in United States Army training exercises to track personnel and unmanned systems during tactical engagements.
- Implemented software for player units and MILES detectors, integrating GPS, NFC, LTE, and BLE to support field-ready tracking, connectivity, and device interaction.

Rockwell Collins Simulated Control Display Unit (SCDU) and Supporting Software (C++ and Rust)

- Developed software for a smart display unit that rendered graphics and text, reported user input over a custom UDP protocol, and supported keypad backlighting and key scanning.
- Implemented serial firmware upgrades using Ymodem and built a cross-platform desktop tool for keypad light balancing, firmware updates, and USB sanitization.

BAE Systems Upgraded Radio Control Unit (uRCU) Software Upgrade (C on AVR)

- Implemented and tested DO-178C requirements for a smart display unit that communicated with a BAE transponder and provided pilot feedback.
- Recovered a legacy codebase without source control by scripting historical version collection and importing the results into Git.

BAE Systems Slim Data Entry Device (DED) (Rust on STM32)

- Developed the main application for a Rust-based STM32 VT100 terminal emulator that rendered serial text and ANSI escape sequences to a display grid.
- Implemented firmware upgrade, touch screen input, and contrast control features, and built a cross-platform desktop test application.

OIL VICTORY Bus Prototype (C++ on Linux)

- Developed a Linux-based touch screen prototype that used VICTORY Bus GPS data to drive an open source mapping application.
- Built a modular VICTORY Bus library and a systemd service to monitor network traffic and forward relevant packets to the mapping stack.

SKILLS

Programming: C, C++, Java, Python, SystemVerilog, VHDL, SystemC, Rust, LaTeX

Technologies: Linux, GitLab, STM32, Nordic, CMake, AVR, GCC, Networking, RTOS, Zephyr